Robotic Technology - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Robotic Technology - NUIST (A35633)

Short Title:	Robot Tech (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Software and Information Systems		
Author	KOMAHONEY		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module focus on introducing the major technology including Robot Operation System(ROS) programming, Visual Inertial Odometry (VIO) principle, sensor fusion based system stats estimation approach ,model-based drone control technology and trajectory planning method which are applied on multirotor helicopter systems.

Programmes

MEng in Electronic Engineering (`WD_EELEN_R`)	/ / E
MEng in Electronic Information Systems (`WD_EELEN_RI`)	3 / 5 / E

Indicative Content

- C/C++: opencv, egin, object programming, matrix operation
- Linux/Ubuntu: git, cmake, toolchain
- Rigid-body Kinematics: Rotations, reflections, and translations, coordinates transformation
- Control Theory: system state space, discrete system, optimum control.
- VIO: geometry, computer vision, nonlinear optimization, camera model and linear algebra
- ROS: launch file, gazebo simulation, mavros, offboard control, ros bag

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Set up Linux-based development environment and cmake-based cross compiler environment.
2. Use ROS to develop mobile robot especially drone control and navigation algorithm.
3. Understand structure of visual inertial odometry algorithm including feature point matching approach of front-end and nonlinear optimization of back-end.
4. Use drone kinematics and model-based control theory such as LQR, driftless flatness to design cascade controller of multirotor type drone, .
5. Apply gazebo simulation to design and turning designed control and navigation algorithm of drone or other mobile robots.

Learning and Teaching Methods

- Delivery of the module will be through a mixture of lectures, tutorial classes and computer laboratory sessions.
- The lectures will develop theory, lead students through worked examples and introduce the context for the module material.
- The tutorial classes will underpin and rehearse the skills demonstrated in the lectures and described module material.

- The practical sessions will utilise ROS and PX4 open source that supports drone control system simulation and design. Students will use this environment to achieve controller and state estimation algorithm covered in lectures.
- Active engagement with frequent practice on examples is strongly encouraged through regular course work and class tests.

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	75%	1,2,3,4
Continuous Assessment	25%	
Practical	35%	5

Assessment Criteria

<40%: Inability to demonstrate knowledge or understanding of the fundamental concepts and techniques in drone kinematic of rate control loop, attitude control loop and velocity control loop. inability to apply such concepts to selected problems.

40%-59%: Able to demonstrate a basic understanding of the fundamental concepts and techniques in drone kinematic and drone control system as concepts outlined in syllabus content.

60%-69%: All the above, in addition be able to use optimization control method to design drone cascade control system with VIO navigation and verify it in ROS.

70%-100%: All the above to an excellent level. Demonstrates an ability to put a solution into a context and assess whether such solutions are meaningful.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- "PX4." www.px4.io
- "ROS." www.ros.org

Requested Resources

- COMPUTER LAB: Networks Lab